

Index Models

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Treynor-Black Model

Index Model

Stock: S

$$\underbrace{r_S - r_F}_{\text{excess return for stock S}} = \underbrace{\alpha_S}_{\text{intercept}} + \underbrace{\beta_S}_{\text{slope}} \times \underbrace{(r_M - r_F)}_{\text{excess return for market}} + \underbrace{\epsilon_S}_{\text{error term}}$$

Wanted! { intercept
slope

$$\underbrace{r_S - r_F}_{\text{Total risk}} = \alpha_S + \underbrace{\beta_S \times (r_M - r_F)}_{\text{systematic risk}} + \underbrace{\epsilon_S}_{\text{idiosyncratic risk}}$$

(A)

$$\text{Total risk} = \text{VAR}(r_S)$$

$$\text{Systematic risk} = \beta_S^2 \cdot \text{VAR}(r_M)$$

$$\text{Idiosyncratic risk} =$$

$$\text{VAR}(r_S) - \beta_S^2 \cdot \text{VAR}(r_M)$$

(B)

Risk premium of the stock:

$$E(r_S) - r_F = \alpha_S + \beta_S (E(r_M) - r_F)$$

(C)

Covariance: Stocks S & U

$$\text{COVAR}(S, U) = ?$$

Apply the index model twice:

$$r_S - r_F = \alpha_S + \beta_S (r_M - r_F) + \varepsilon_S$$

$$r_U - r_F = \alpha_U + \beta_U (r_M - r_F) + \varepsilon_U$$

$$\text{COVAR}(S, U) = \beta_S \times \beta_U \times \text{VAR}(r_M)$$

Treynor - Block : Take the risk premium + COVAR formulas from above $\Rightarrow$ use them to find the Optimal Risky Portfolio (ORP)



Treynor - Block found analytical formulas for the weights of ORP



They found that ORP is going to be a mix of:

① The market, $M = \boxed{\text{passive}}$

② An active portfolio, $A = \boxed{\text{active}}$